

FEASIBILITY STUDIES AND AB ENGINEERING ECONOMY

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- a. Free
- b. Scarce
- c. Unlimited
- d. Restricted

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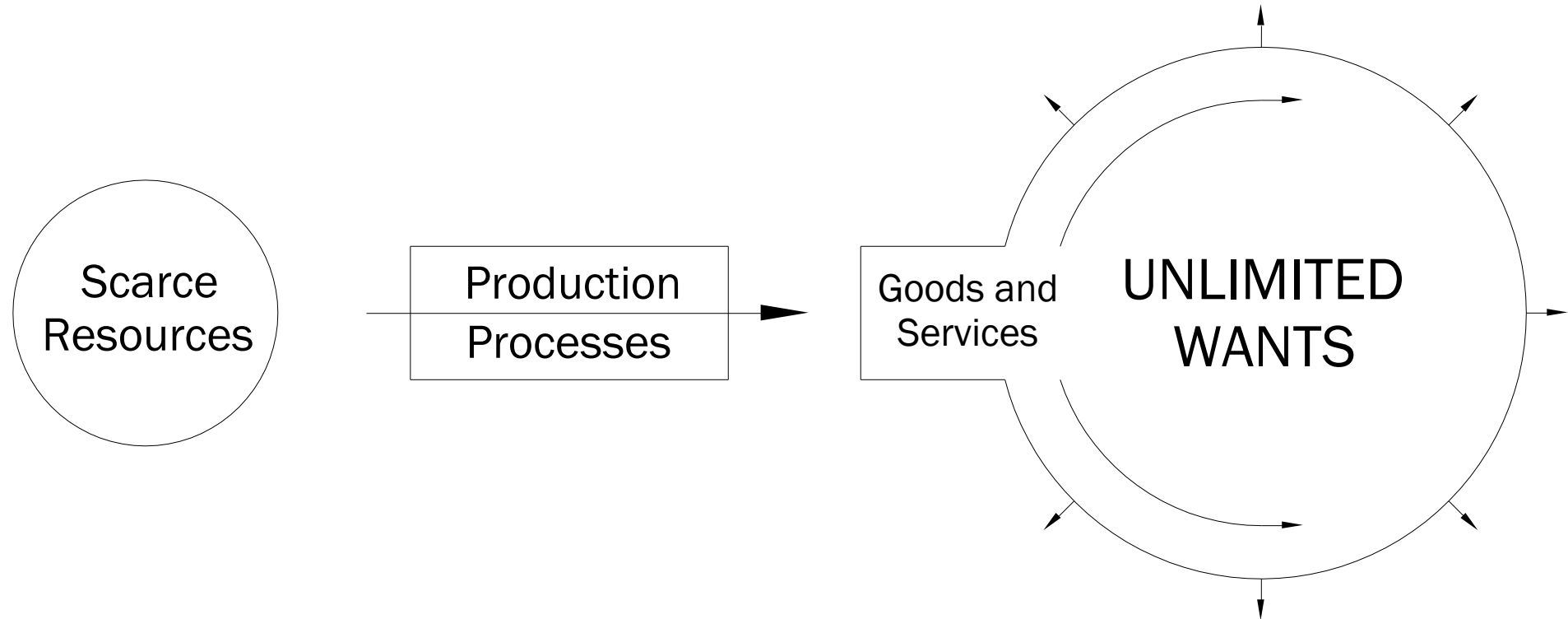
c. Unlimited

d. Restricted

ECONOMICS

- The branch of knowledge concerned with the production, consumption, and transfer of wealth.
- The proper allocation and utilization of resources, especially available scarce resources to maximize their benefits for the satisfaction of people.
- This is a subject concerned with the optimization of available resources, in an efficient manner.

Why Study Economics?



SCARCITY AND CHOICE

Scarcity is the main obstacle which economics aims to hurdle. Because of scarcity, there is a need to decide how to allocate resources and make choices on the goods to be produced.

Opportunity Cost - refers to the value of the best forgone opportunity.

2. The objective of a rational consumer is to
- a. buy as much as his income allows of the cheapest articles
 - b. avoid purchasing the most expensive commodities
 - c. secure the highest level of satisfaction from his money income
 - d. spread his expenditures over as many products as possible

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BASIC NOTIONS IN ECONOMICS

- **Allocation** – of putting resources (inputs, raw materials, and inputs of production) or products (output or good) to some use.
- **Scarcity** – that there simply not enough inputs to make enough goods to satisfy the nearly invisible wants of all individuals and groups in the world
- The **idea of establishing goals, ends or objectives**. Human wants are unlimited, which means that wants compete for the scarce resources.
- Finally, we must include some **reference to time**. This permits flexibility between short and long time periods and allows for changes in the entire system of goals over time

3. Which of the following best describes the factor of production known as "land"?

- A. All man-made tools and buildings
- B. The physical and mental effort of people
- C. Natural resources used in production
- D. The risk-taking and innovation of business owners

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- B. Labor
- C. Capital
- D. Entrepreneurship

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FACTORS OF PRODUCTION

LABOR

- the work done by people. The value of the workforce depends on workers' education, skills, and motivation.

Unskilled labor

- Those are labors that requires only human effort. You don't need to be educated to be an unskilled worker.

Semi-skilled Labor

- Requires education but it does not require advance training.

Skilled Labor

- Requires trainings, and a specialized know-how.

Professional Labor

- It requires a very specialized experience.

FACTORS OF PRODUCTION

LAND

- It includes raw property and anything that comes from the ground.

CAPITAL

- These are man-made objects like machinery, equipment, and chemicals that are used in production.

ENTREPRENEURSHIP

- the drive to develop an idea into a business. An entrepreneur combines the other three factors of production to add to supply. The most successful are innovative risk-takers.

EMPLOYMENT CATEGORIES

- 1. Primary Employment** – people employed in extractive industries of farming, forestry, fishing, and mining
- 2. Secondary Employment** – those people having jobs in the manufacturing industries, processing or fabricating materials into semi-finished and finished goods
- 3. Tertiary Employment** – people whose job is to serve the needs of the other two employment sectors, as well as to provide much of the maintenance of the total economy (finance, education, transportation, construction, government, etc)

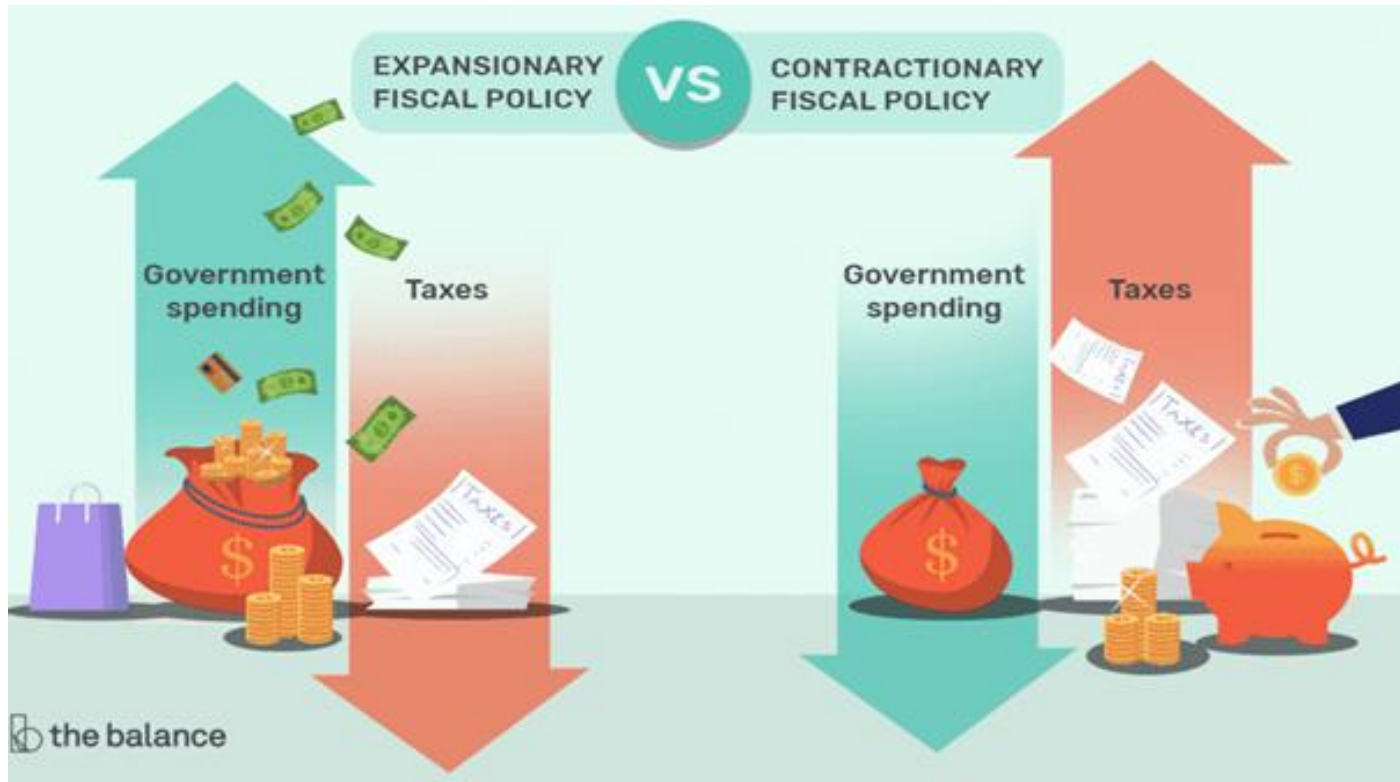
5. What is the primary goal of contractionary monetary policy?

- A. To reduce inflation
- B. To reduce unemployment
- C. To increase consumer spending
- D. To stimulate economic growth

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ECONOMIC POLICIES



An **expansionary policy** increases the supply of money in an economy.

A **contractionary policy** decreases the supply of a country's currency.

6. If the price of corn rises and farmers plant more corn, this is an illustration of:

- A. The Law of Demand
- B. The Law of Supply
- C. Elasticity of demand
- D. The Law of Diminishing Marginal Utility

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LAW OF DEMAND

Price – is the value of the product or service expressed in monetary units.

Demand

Quantity of goods & services that a consumer is willing to buy at a given price & time

Elasticity

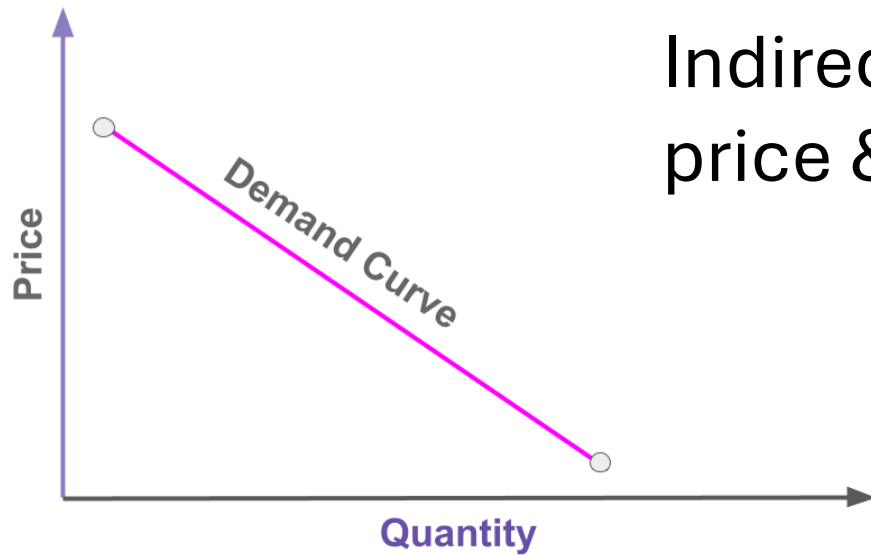
The degree to which a demand or supply curve reacts to a change in price and other determinants.

LAW OF DEMAND

“As the price increases, the quantity demanded decreases; and as price decreases, quantity demanded increase.” This is only true if the assumption of “ceteris paribus” (all things are equal or constant) is applied.”

Indirect and negative relationship of price & quantity demanded

if P increases, D decreases
if P decreases, D increases



LAW OF DEMAND

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Income Effects: The same amount of money can buy more goods when prices are lower, but lesser goods when prices are higher.

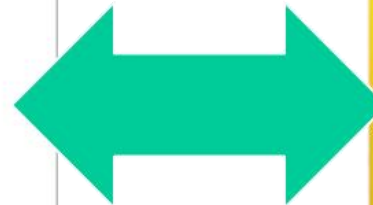
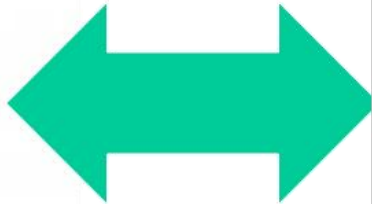
Substitution Effect: Consumers tend to buy goods with lower prices.

A **normal good** sees an increase in demand when incomes rise. Demand for **inferior goods** increases as income falls.

LAW OF DEMAND

Substitutes

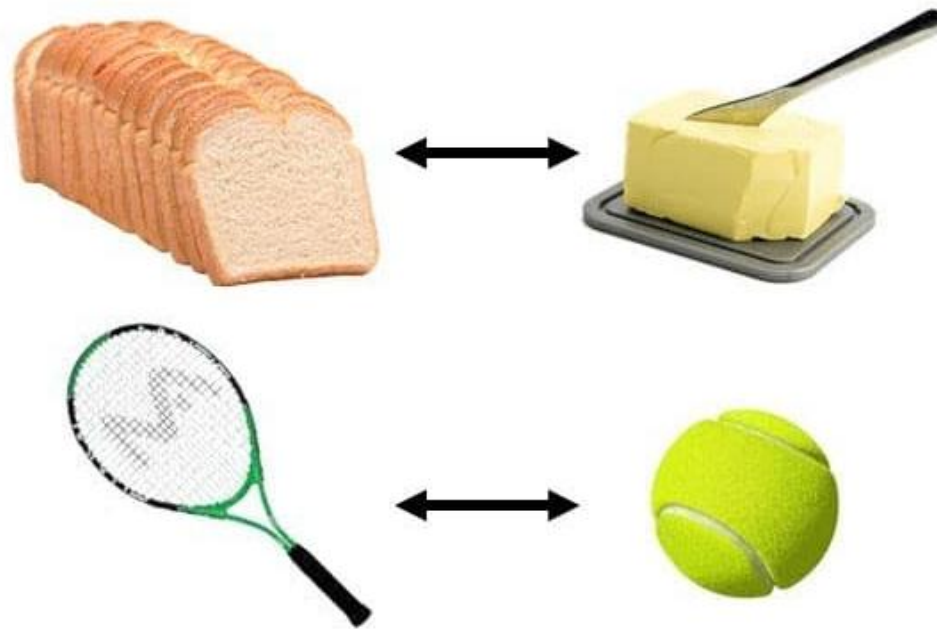
two goods for which an increase in the price of one leads to an increase in the demand for the other.



LAW OF DEMAND

Complements

two goods for which an increase in the price of one leads to a decrease in the demand for the other



LAW OF DEMAND

The Income Effect

When the price of a product decreases, *ceteris paribus*, consumers have more relative income. They can, therefore, purchase additional products (and vice versa).



LAW OF DEMAND

Demand Schedule – a tabular presentation showing the price and quantity demanded for a particular good.

Demand Curve – is the curve that shows the relationship between price and quantity demanded.

Demand Function – a mathematical model showing the relationship between price and the quantity demanded.

Price	Quantity Demanded per Month
\$15	1
12	2
9	3
6	5
3	7



7. The term “ceteris paribus” means
- a. Other things held constant
 - b. Absence of government intervention
 - c. Scarcity or limited
 - d. Closed economy

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- a. Normal good
- b. Inferior good
- c. Substitute good
- d. Complementary good

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9. The quantity of coffee bought in the market is likely to increase when:
- a. The price of tea increases
 - b. The price of coffee increases
 - c. Coffee is linked to cancer by new research
 - d. The general income level of the population drops

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10. When an individual's income rises (other things constant) the demand for a normal good

a. rises

b. remains the same

c. falls

d. we cannot say without additional info

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11. Studies revealed that margarine is an inferior good.

Therefore, an increase income will result in:

- a. An increase in demand for margarine
- b. A shift to the left of the demand curve for margarine
- c. A shortage of margarine
- d. An increase in the supply of margarine

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12. Market research has determined that during hard economic times, sales of local canned sardines improve, This indicates that:

- a. Local canned sardines are inferior goods
- b. Canned sardines companies have effective advertising campaigns
- c. Imported sardines are bought only by rich people
- d. Petroleum prices have no effect on the market for sardines

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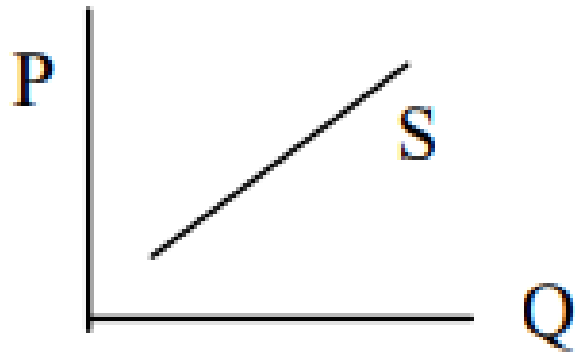
LAW OF SUPPLY

Supply - willingness of the firm to produce/sell goods at different price levels.

- The quantity of goods and services that the sellers are willing and able to sell or produce.
- A reflection of the choice of the seller, the producer of the goods and services being brought by the households in the market.

LAW OF SUPPLY

“As the price increases, quantity supplied also increases; and as price decreases, quantity supplied also decreases.” This is only true if the assumption of “ceteris paribus” is applied.



Direct and positive relationship of price
& quantity demanded

if P increases, S increases
if P decreases, S decreases

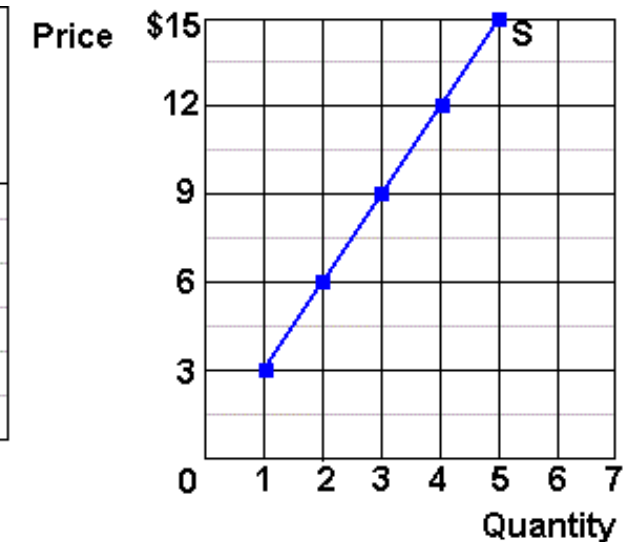
LAW OF SUPPLY

Supply Schedule – tabular presentation of supply showing the price and quantity supplied

Supply Curve – the graphical presentation of the supply schedule.

Supply function – the mathematical model showing the relationship between price and quantity supplied.

Price	Quantity Supplied per Month
\$15	5,000
12	4,000
9	3,000
6	2,000
3	1,000



LAW OF SUPPLY AND DEMAND

when true competition exists, the price of a product will be that value where supply is equal to the demand.

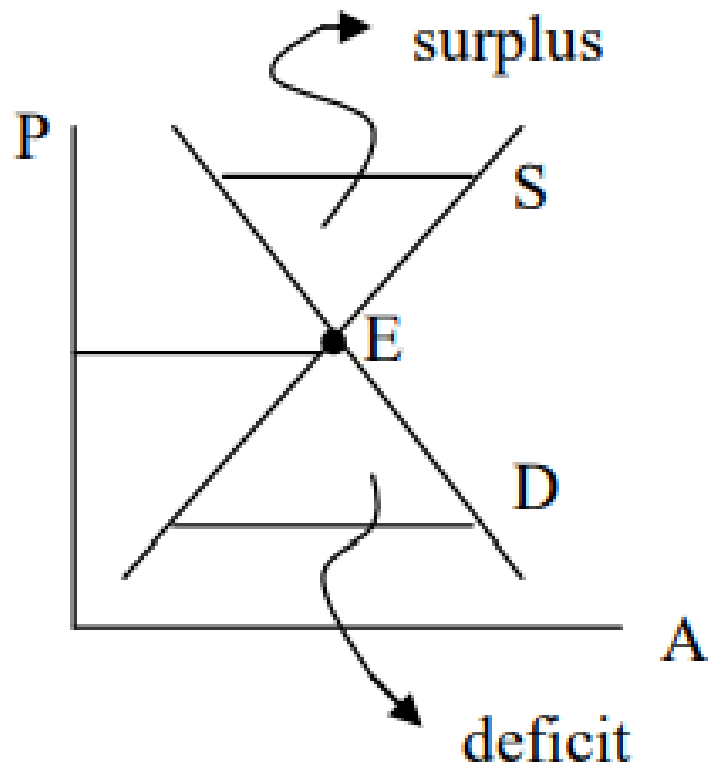
Market Equilibrium – The point in the where the demand and supply meet. exchange takes place; consumers and sellers are satisfied.

Equilibrium Price – the price at which the quantity demanded is equal to the quantity supplied

Equilibrium Quantity – the quantity where demand and supply are equal.

LAW OF SUPPLY AND DEMAND

Equilibrium $D = S$



If $D > S$
ceiling
If $D < S$
floor

shortage; P increases = price

surplus; P decreases = price

KINDS OF PRICE CONTROL

Price floor- the minimum price; bias towards suppliers

Price ceiling- the maximum price that can be imposed; bias towards the consumers

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- a. shortage
- b. surplus
- c. taxes
- d. tariffs

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14. Which of the following statements is incorrect?

- a. The law of supply yields a negatively sloped supply curved
- b. The law of demand yields a negatively sloped curved
- c. A shortage or surplus is observed if disequilibrium occurs in the market
- d. none of the above

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15. The Engel Curve shows the relationship between:

- a. Income and price
- b. Income and quantity demanded of a good
- c. Price and quantity supplied
- d. Income and production level

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ECONOMIC THEORIES

Maslow's Hierarchy of Needs - The theory states that humans are motivated to fulfill their needs in a hierarchical order.

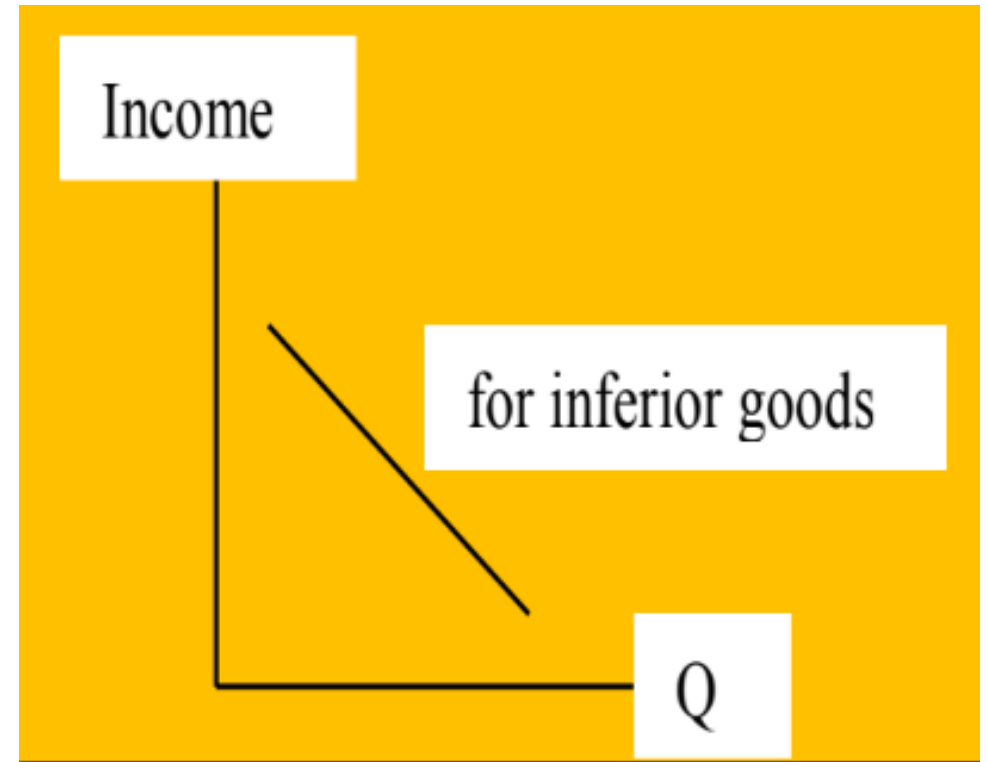
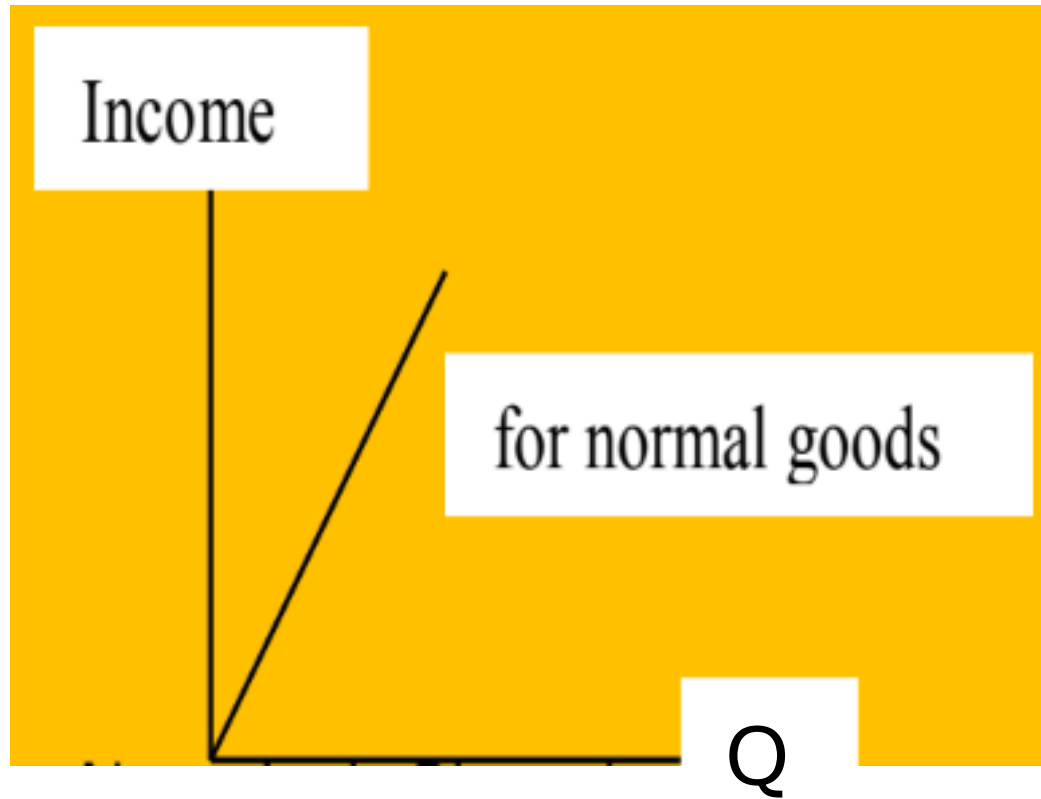
Resource-Use Choice Model - Until the basic needs are met, there is a “trade-off” between resource use–waste management and environmental degradation and maintaining the pure state of the natural environment; based on the idea of scarcity.

Engel's Law –states that the percentage of income allocated for food purchases decreases as income rises. As a household's income increases, the percentage of income spent on food decreases while the proportion spent on other goods (such as luxury goods) increases.



Maslow's hierarchy of needs

ECONOMIC THEORIES



16. According to Engel's Law, which of the following households is most likely to spend the highest percentage of income on food?
- a. High-income urban household
 - b. Low-income rural household
 - c. Middle-income suburban household
 - d. High-income rural household

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 - b. Low-income rural household**
 - c. Middle-income suburban household
 - d. High-income rural household

ECONOMIC THEORIES

Law of Diminishing Utility: “An increase in the quantity of any good consumed or acquired by an individual will decrease the amount of satisfaction derived from good.”

Utility – the capacity of a commodity to satisfy human want; if the utility of a certain good is great, the demand for that good will be great.

Marginal Utility – utility of the last unit of the same commodity which is consumed or acquired.

Total utility – total satisfaction in the consumption goods & services

Marginal Unit – the last unit of similar commodities consumed or acquired

ECONOMIC THEORIES



Very thirsty
High marginal benefit



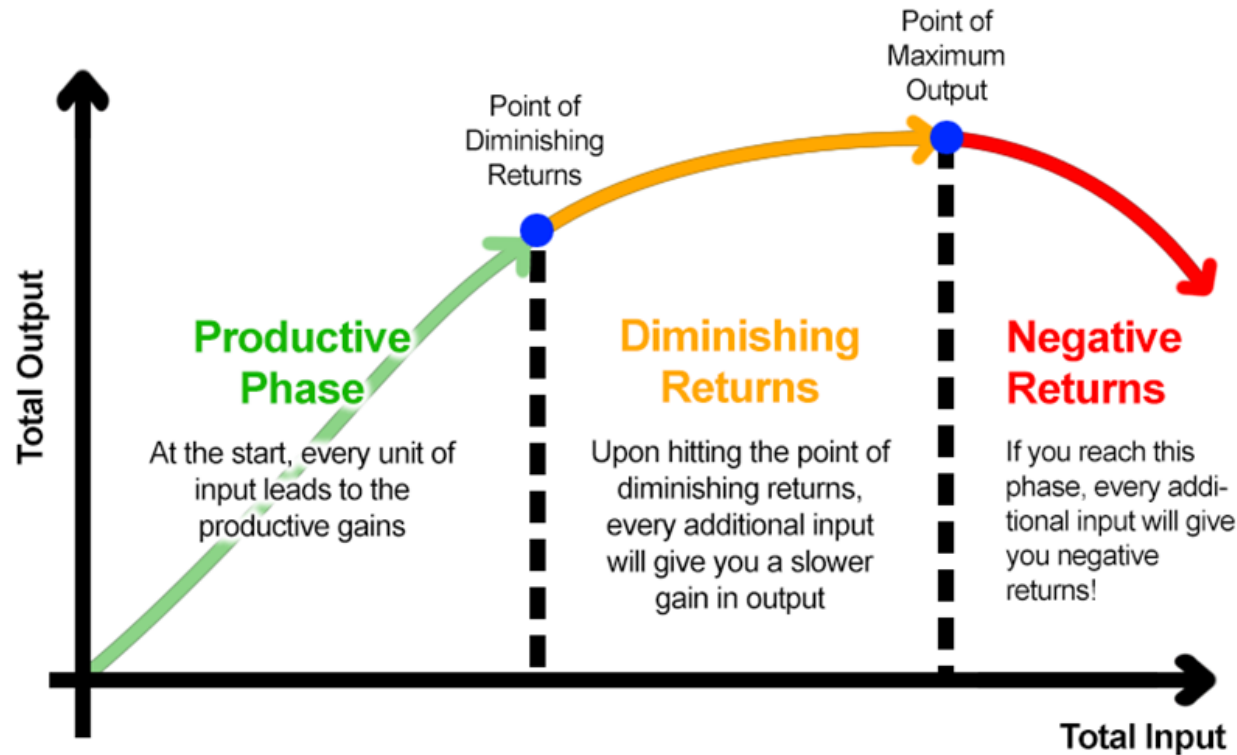
Moderately thirsty
Moderate marginal benefit



Not thirsty at all
Low marginal benefit

ECONOMIC THEORIES

Law of Diminishing Returns - when one of the factors of production is fixed in quantity or is difficult to increase, increasing the other factors of production will result in a less than proportionate increase in output.



17. The satisfaction that a person receives from the consumption of goods is called

- a. Budget constraints
- b. Income
- c. Consumer surplus
- d. Utility

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AB ENGINEERING ECONOMY

Engineering Economics

- The study of the cost factors involved in engineering projects, and usually the results of such study in employing the most efficient cost-saving techniques without affecting the safety and soundness of the project.
- Involves formulating, estimating, and evaluating the expected economics outcomes of alternatives designed to accomplish a defined purpose

AB ENGINEERING ECONOMY

Cash Flow Diagramming

Cash flows - are the amounts of money estimated for future projects or observed for project events that have taken place

- **Cash inflows** - are the receipts, revenues, incomes, and savings generated by project and business activity
- **Cash outflows** - are costs, disbursements, expenses, and taxes caused by projects and business.
- **Net cash flow** = cash inflows - cash outflows

INTEREST

Interest- money paid for the use of borrowed money

Simple Interest- calculated using the principal only, ignoring any interest accrued in preceding interest periods

Ordinary simple interest – computed on the basis of 12 months of 30 days each or 360 days a year.

Exact simple interest – based on the exact number of days in a year, 365 days for an ordinary year and 366 days for a leap year.

$$I = P i n$$

$$F = P (1+in)$$

Where: F = accumulated amount or future worth

P = principal or present worth

i = interest rate per interest period

n = number of interest period

SIMPLE INTEREST

Determine the ordinary simple interest on Php700 for 8 months and 15 days if the rate of interest is 15%.

SIMPLE INTEREST

A loan of Php5,000 at an interest rate of 12% per annum for the period of February to May. Determine the ordinary simple interest due in May.

SIMPLE INTEREST

Determine the exact simple interest on Php1000.00 for the period of February 5 to October 21, 2024, at 15% interest.

SIMPLE INTEREST

Determine the accumulated amount that would be due at the end of 18 months for Php15,000.00 borrowed at simple interest at a rate of 6% per annum.

SIMPLE INTEREST

18. What will be the future worth of money after 12 months, if the sum of P25, 000 is today at simple interest rate of 1% per month?

- A. P30, 000
- B. P29, 000
- C. P28, 000
- D. P27, 859

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C. P28, 000

D. P27, 859

SIMPLE INTEREST

19. P4000 is borrowed for 71 days at 16% per annum simple interest. How much will be due at the end of 75 days?

- A. P4, 133.33
- B. P4, 333.33
- C. P4, 166.67
- D. P4, 150.00

SIMPLE INTEREST

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B. P4, 333.33

C. P4, 166.67

D. P4, 150.00

SIMPLE INTEREST

20. A deposit of P110, 000 was made for 31 days. The net interest after deducting 20% withholding tax is P890.36. Find the rate of return annually.

- A. 11.95%
- B. 12.75%
- C. 11.75%
- D. 12.25%

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INTEREST

COMPOUND INTEREST- the interest accrued for each interest period is calculated on the principal plus the total amount of interest accumulated in all previous periods

Discrete Compounding – the interest is compounded at the end of each finite length period, such as a month, a quarter or a year

$$F = P (1 + i)^n$$

Continuous Compounding – it is assumed that cash payments occur once a year, but the compounding is continuous throughout the year

$$F = Pe^{rn}$$

COMPOUND INTEREST

Sandy, a manufacturing engineer, just received a year-end bonus of ₱10,000 that will be invested immediately. With the expectation of earning at the rate of 8% per year, Sandy hopes to take the entire amount out in exactly 20 years to pay for a family vacation when the oldest daughter is due to graduate from college. Find the amount of funds that will be available in 20 years.

COMPOUND INTEREST

The XXX factory will require an investment of ₱200 million to construct. Delays beyond the anticipated implementation year of 2012 will require additional money to construct the factory.

Assuming that the cost of money is 10% per year, compound interest, determine the following.

- (a) The equivalent investment needed if the plant is built in 2015.
- (b) The equivalent investment needed had the plant been constructed in the year 2008.

COMPOUND INTEREST

A sum of Php1,000 is invested now and left for eight years, at which time the principal is withdrawn. The interest has accrued is left for another eight years. If the effective annual interest rate is 5%, what will be the withdrawal amount at the end of the 16th year?

COMPOUND INTEREST

What is the present worth of two P100 payments at the end of the third year and fourth year? The annual interest rate is 8%.

- A. P153
- B. P160
- C. P162
- D. P127

COMPOUND INTEREST

What is the present worth of two P100 payments at the end of the third year and fourth year? The annual interest rate is 8%.

A. P153

B. P160

C. P162

D. P127

COMPOUND INTEREST

On his 6th birthday, a boy is left an inheritance. the inheritance will be paid in a lump sum of P10, 000 on his 21st birthday. What is the present value of the inheritance as of the boy's 6th birthday, if the interest is compounded annually? Assume $i=4\%$

- A. P6, 500
- B. P8, 600
- C. P5, 500
- D. P7, 500

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- A. P6, 500
- B. P8, 600
- C. P5, 500
- D. P7, 500

NOMINAL AND EFFECTIVE INTEREST RATES

the interest accrued for each interest period is calculated on the principal plus the total amount of interest accumulated in all previous periods

Nominal rate of interest – specifies the rate of interest and a number of interest periods in one year.

$$i = \frac{r}{m}$$

$$F = p \left(1 + \frac{r}{m} \right)^{mn}$$

Effective rate of interest – the actual or exact rate of interest on the principal during one year

$$i_e = (1 + i)^m - 1$$

$$i_e = e^r - 1$$

NOMINAL AND EFFECTIVE INTEREST RATES

Calculate the effective rate corresponding to each of the following rates:

- (a) 9% compounded semi-annually,
- (b) 9% compounded quarterly,
- (c) 9% compounded bi-monthly,
- (d) 9% compounded monthly, and
- (e) 9% compounded continuously.

NOMINAL AND EFFECTIVE INTEREST RATES

How many years are required for Php1,000.00 to increase to Php2,000.00 if invested at 9% per year:

- (a) Compounded annually,
- (b) Compounded Semiannually,
- (c) Compounded Quarterly,
- (d) Compounded Monthly,
- (e) Compounded Daily, and
- (f) Compounded continuously.

NOMINAL AND EFFECTIVE INTEREST RATES

What is the corresponding effective rate of 18% compounded semi-quarterly?

- A. 19.25%
- B. 19.48%
- C. 18.46%
- D. 18.95%

NOMINAL AND EFFECTIVE INTEREST RATES

What is the corresponding effective rate of 18% compounded semi-quarterly?

- A. 19.25%
- B. 19.48%**
- C. 18.46%
- D. 18.95%

NOMINAL AND EFFECTIVE INTEREST RATES

The effective rate of 14% compounded semi annually is

- A. 14.49%
- B. 12.36%
- C. 14.94%
- D. 14.88%

NOMINAL AND EFFECTIVE INTEREST RATES

The effective rate of 14% compounded semi annually is

- A. 14.49%
- B. 12.36%
- C. 14.94%
- D. 14.88%

DISCOUNT AND INFLATION RATE

Discount rate – an estimate of how much less something is worth if it is received in the future

$$d = \frac{i}{1 + i} \quad i = \frac{d}{1 - d}$$

Inflation rate- an increase in the amount of money necessary to obtain the same amount of goods or services before the inflated price was present.

$$i_f = i + f + if$$

Purchasing power, or buying power, measures the value of a currency in terms of the quantity and quality of goods or services that one unit of money will purchase.

Deflation is the opposite of inflation in that when deflation is present, the purchasing power of the monetary unit is greater in the future than at present.

DISCOUNT AND INFLATION RATE

A farmer borrows ₱10,000.00 from a farmer's cooperative. The interest is to be deducted from the loan at the time the money is borrowed at 15% interest rate (simple). He has to pay back ₱10,000 at the end of one year. What is actual rate of interest?

DISCOUNT AND INFLATION RATE

AMS wants to determine whether it should upgrade a piece of equipment now or later. If the company selects plan A, the upgrade will be purchased now for Php200,000. However, if the company selects plan I, the purchase will be deferred for 3 years when the cost is expected to rise to Php300,000. It expects a real MARR of 12% per year. The inflation rate has averaged 3% per year. From only an economic perspective, determine whether the company should purchase now or later (a) when inflation is not considered a (b) when inflation is considered.

Which situation is better and attainable for a good economy?

- A. High discount rate
- B. Low discount rate
- C. Fast discount rate
- D. No discount rate

Which situation is better and attainable for a good economy?

A. High discount rate

B. Low discount rate

C. Fast discount rate

D. No discount rate

ANNUITY

ANNUITY is a series of equal payments occurring at equal periods of time

Occurs in the following instances:

- payment of a debt by a series of equal payments at equal intervals of time.
- accumulation of a certain amount by setting equal amounts periodically.
- substitution of a series of equal amounts periodically in lieu of a lump sum at retirement of an individual.

ANNUITY

Notation	Factor Name	Find/Given	Factor Formula	Standard Notation Equation
$(P/A, i, n)$	Uniform series present worth	P/A	$\frac{(1 + i)^n - 1}{i(1 + i)^n}$	$P = A(P/A, i, n)$
$(A/P, i, n)$	Capital recovery	A/P	$\frac{i(1 + i)^n}{(1 + i)^n - 1}$	$A = P(A/P, i, n)$
$(F/A, i, n)$	Uniform series compound amount	F/A	$\frac{(1 + i)^n - 1}{i}$	$F = A(F/A, i, n)$
$(A/F, i, n)$	Sinking fund	A/F	$\frac{i}{(1 + i)^n - 1}$	$A = F(A/F, i, n)$

ANNUITY

Uniform Series Present Worth

$(P/A, i, n)$ - Converts a uniform series of future payments into its present worth.

Capital Recovery

$(A/P, i, n)$ - Converts a present value into a series of equal payments

Sinking Fund

$(A/F, i, n)$ - Finds equal payments needed to accumulate a future amount.

Uniform Series Compound Amount

$(F/A, i, n)$ - Converts a series of equal payments into a future value

ORDINARY ANNUITY

One where the equal payments are made at the end of each payment period starting from the first period

Four essential elements

1. The amounts of all payments are equal.
2. The payments are made at equal intervals of time.
3. The first payment is made at the end of the first period and all payments thereafter are made at the end of the corresponding period.
4. Compound interest is paid on all amounts in the annuity.

ORDINARY ANNUITY

A man paid 10% down payment of P200,000.00 for the house and lot and agreed to pay the 90% balance on monthly installments for 60 months at an interest rate of 15% compounded monthly. Compute the amount of the monthly payments.

ORDINARY ANNUITY

What is the present worth of a 3 years annuity paying P3,000.00 at the end of each year, with interest at 8% compounded annually?

ORDINARY ANNUITY

What is the accumulated amount of the 5 year annually paying P6,000.00 at the end of each year, with interest at 15% compounded annually?

ORDINARY ANNUITY

What is the future worth of P600 deposited at the end of every 3 months for 4 years if the interest rate is 12% compounded quarterly?

\$100K today or in a year?

Would you rather receive \$100,000 today or \$100,000 one year from now?

Today



\$100,000

OR

One Year



\$100,000

- a) \$100,000 today
- b) \$100,000 in one year
- c) It's the same amount, so it doesn't matter
- d) It depends

\$100K today or in a year?

You should prefer \$100,000 today.



- a) **\$100,000 today**
- b) \$100,000 in one year
- c) It's the same amount, so it doesn't matter
- d) It depends

\$100K today or in a year?

This is because a dollar today can be invested and earns interest.



- It's better to have a dollar today than in the future. You can invest it and get more than a dollar in the future.
- **A dollar today is worth more than a dollar in the future.**

The time value of money

A dollar today is more valuable than a dollar later.

This is called the **time value of money**...



PRESENT VALUE

Money now is more valuable than money later on.

Why? Because you can use money to make more money!

Present Value Formula

$$PV = FV \frac{1}{(1 + r)^n}$$

FV = Future Value

r = rate of return

n = number of periods

PRESENT VALUE

Colet promised you to pay Php 20,000 in 4 years time. What is the Present Value (using at 10% rate of discount)?

FEASIBILITY STUDY DESIGN

Feasibility Study

- an exploratory and evaluation tool used to determine whether an intended venture or business idea is viable/profitable or not
- the evaluation is done based on past and existing information that are projected into the future. The resulting evaluation will serve as a basis of deciding whether the proposed idea is to be pursued or not.

FEASIBILITY STUDY DESIGN

Project Feasibility Study - an analytical report of the assessment of a project concerning its market, technical, financial, socio-economic, and management feasibilities. It is the culmination report of all the preparatory work that provides a comprehensive review of all aspects of the project before a final decision about its implementation is taken. A standard feasibility study (FS) contains the following parts which provide the basis for evaluation: (a) project summary, (b) market feasibility, (c) technical feasibility, (d) financial feasibility, (e) socio-economic feasibility, and (f) management feasibility.

Source: DA MO No. 61 Series of 2020

https://www.da.gov.ph/wp-content/uploads/2020/12/mo61_s2020.pdf

FEASIBILITY STUDY DESIGN

Advantages in Preparing Feasibility Study

1. It offers means of determining whether valuable resources may be committed in an undertaking
2. It reduces the potential risk of failure for an undertaking
3. It allows for the identification of critical issues and critical resources in an undertaking

Disadvantages in Preparing Feasibility Study

1. It requires technical knowledge
2. It requires initial expenses
3. It requires time and effort

5 FEASIBILITY INDICATORS for both financial and economic analyses in Feasibility Studies, ranked by importance:

1. Net Present Value (NPV)
2. Internal Rate of Return (IRR)
3. Benefit-Cost Ratio (BCR)
4. Return on Investment (ROI)
5. Payback Period

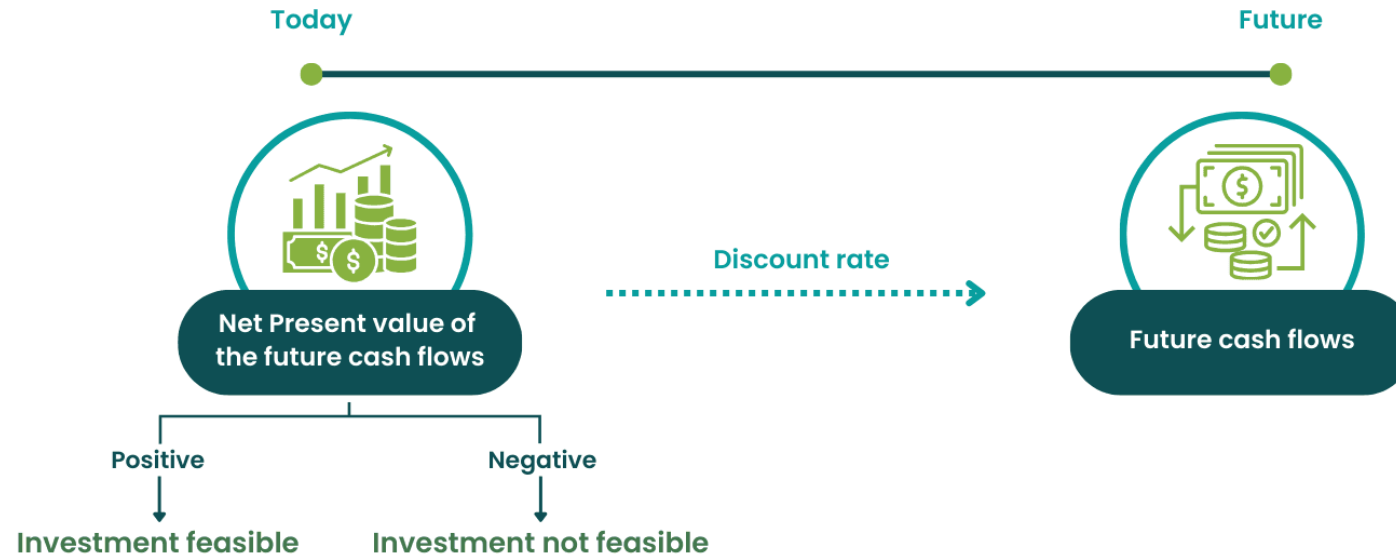
PRESENT WORTH/NPV

The NPV of a project or investment reflects the degree to which cash inflow, or revenue, equals or exceeds the amount of investment capital required to fund it.

STEPS TO CALCULATION NPV

1. Determine your initial investment
2. Determine a time period to analyze.
3. Estimate your cash flows for each time period
4. Determine the appropriate discount rate.
5. Discount your cash flows.

The **discount rate** refers to the rate of interest that is applied to future cash flows of an investment to calculate its present value. It is the rate of return that companies or investors expect on their investment.



When the interest rate increases, the discount rate used in the NPV calculation also increases. This higher discount rate reduces the present value of future cash inflows, leading to a lower NPV.

PRESENT WORTH/NPV

Based on the concept of equivalent worth of all cash flows relative to some base or beginning point in time called present

Criterion:

- For project Evaluation: A project is acceptable if $PW \geq 0$
- For Mutually Exclusive Alternative Selection: Select the alternative with greater PW
- For One or More Independent projects: Select all projects with $PW \geq 0$ at the MARR.

Minimum Attractive Rate of Return (MARR) is a reasonable rate of return established for the evaluation and selection of alternatives

PRESENT WORTH/NPV

Sheena needs Php 1,000 now and will pay you back Php 1,350 in a year. Is that a good investment when you can get at 10% per year?

PRESENT WORTH/NPV

Invest Php 20,000 now and receive 3 yearly payments of Php 5,000 each, plus Php 12,000 in the 3rd year. Use 10% discount Rate. Find NPV.

PRESENT WORTH/NPV

A project with a 3 year life and a cost of Php 100,000 generates revenue of Php 25,000 in year 1, Php 45,000 in year 2, and Php 65, 000 in year 3. If the discount rate is 8%, what is the NPV of the project?

PRESENT WORTH/NPV

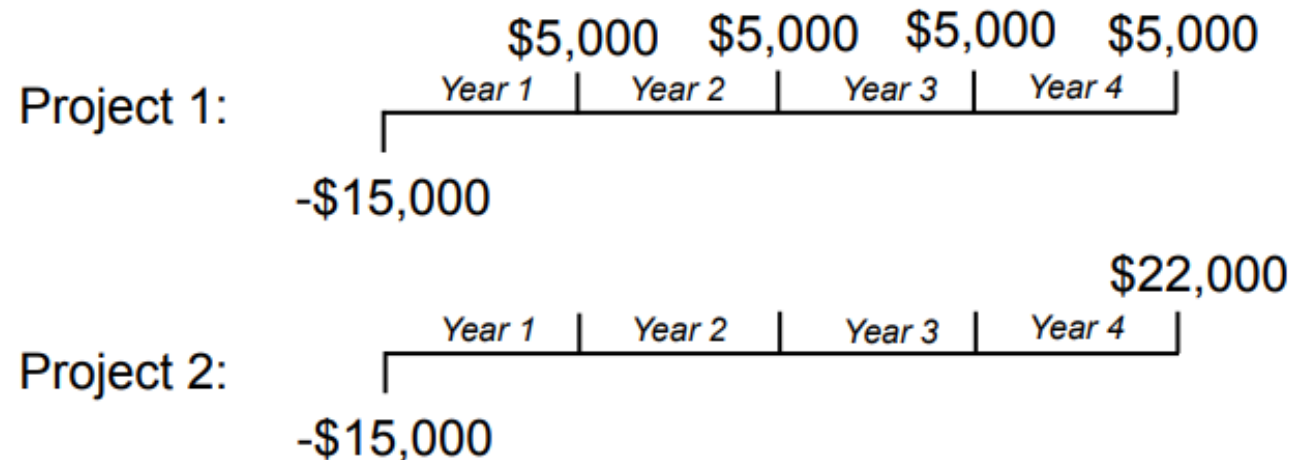
A project with a 4-year life and a cost of Php 225,000 generates revenue of Php 48,000 in year 1, Php 67,000 in year 2, Php 95, 000 in year 3 and Php 110,000 in year 4. If the discount rate is 15%, Can be accepted the project?

PRESENT WORTH/NPV

Michael wins the state lottery with a Php250,000 jackpot. To receive the full jackpot, he must wait five years. If he collects his winnings today, he will only receive Php200,000. If the interest rate is 6%, what is the better deal?

PRESENT WORTH/NPV

Consider two business projects, each of which require an initial investment of \$15,000 and each of which are valued with an interest rate of 11%. The first project generates \$5,000 cash flows in years one to four and the second project generates a single cash flow of \$22,000 in four years. Which project, if any, should be undertaken for \$15,000?



FUTURE WORTH

All cash inflows and outflows are compounded forward to a reference point in time called the future

Criterion:

- For project Evaluation: A project is acceptable if $FW \geq 0$
- For Mutually Exclusive Alternative Selection: Select the alternative with greater FW
- For One or More Independent projects: Select all projects with $FW \geq 0$ at the MARR.

FUTURE WORTH

An agricultural cooperative is evaluating whether to purchase a solar-powered irrigation system. The cost of the system is ₱100,000 today. It will save the cooperative ₱25,000 annually in diesel fuel costs for the next 5 years. If the interest rate is 12%, is the project economically feasible based on Future Worth analysis?

ANNUAL WORTH AND CAPITAL RECOVERY

The annual equivalent revenue (or savings) minus annual equivalent expenses, less its annual equivalent capital recovery cost

Criterion:

- For project Evaluation: A project is acceptable if $AW \geq 0$
- For Mutually Exclusive Alternative Selection: Select the alternative with greater AW
- For One or More Independent projects: Select all projects with $AW \geq 0$ at the MARR.

ANNUAL WORTH AND CAPITAL RECOVERY

Compare the alternatives shown below on the basis of a future worth analysis, using an interest rate of 8% per year.

	P	Q
First cost, Php	-23,000	-30,000
Annual Operating Cost, Php	-4,000	-2,500
Salvage Value, Php	3000	1000
Life years	3	6

INTERNAL RATE OF RETURN

The rate paid on the unpaid balance of borrowed money, or the rate earned on the unrecovered balance of an investment, so that the final payment or receipt brings the balance to exactly zero with interest considered.

Decision Rule

- Single projects: $IRR \geq MARR$
- For mutually Exclusive alternative: $IRR(\text{Incremental}) \geq MARR$

Internal Rate of Return is the interest rate that makes the Net Present Value Zero.

INTERNAL RATE OF RETURN

An entrepreneur is deciding whether to expand her business. Doing so would require an investment of Php300,000 and would generate an extra profit of Php200,000 a year for the next four years. Calculate the IRR for this investment. If the entrepreneur requires that the return on the investment exceed 20%, will she undertake this investment?

BENEFIT/COST RATIO

Ratio of present value of benefits to present value of costs.

- Costs—estimated expenditures to the government entity for construction, operation, and maintenance of the project, less any expected salvage value.
- Benefits—advantages to be experienced by the owners, the public.
- Disbenefits—expected undesirable or negative consequences to the owners if the alternative is implemented. Disbenefits may be indirect economic disadvantages of the alternative.

If $BCR > 1$, project is economically justified.

BENEFIT/COST RATIO

The Parks and Recreation Department has estimated that the initial cost of a “barebones” permanent river park will be Php2.3 million. Annual upkeep costs are estimated at Php120,000. Benefits of Php340,000 per year and disbenefits of Php40,000 per year have also been identified. Using a discount rate of 6% per year, calculate (a) the conventional B/C ratio.

RETURN OF INVESTMENT

The financial ratio used to calculate the benefit an investor will receive in relation to their investment cost.

$$\text{ROI} = \text{Net profit} / \text{Total investment, expressed as a percentage}$$

RETURN OF INVESTMENT

An agricultural engineer invests ₱120,000 in a small rice milling machine. After one year, it generates a net income of ₱30,000. What is the ROI?

PAYBACK PERIOD

- It is used to determine the amount of time, usually expressed in years, required to recover the first cost of an asset or project.
- An estimated time for the revenues, savings, and any other monetary benefits to completely recover the initial investment plus a stated rate of return i .

PAYBACK PERIOD

A small rice thresher costs ₱90,000. It generates ₱30,000 per year in net cash savings. What is the payback period?

PAYBACK PERIOD

Determine the payback period for the following farm investment

Year	Cash flow
0	-50,000
1	10,000
2	12,000
3	15,000
4	18,000
5	20,000

BREAK-EVEN ANALYSIS

- **Break-even Point (for one decision variable)** - Level of production when the total income is equal to total expense

$$R = C_T \quad 1$$

$$R = pD \quad 2$$

$$C_T = VC_T + FC \quad 3$$

$$VC_T = (VC)(D) \quad 4$$

Fixed Cost – those that are not affected by changes in activity level over a feasible range of operations for the available capacity.

Variable Cost – those associated with an operation that will vary in total with the quantity of output.

Revenue (R) – total amount of income generated by the sales of goods or services.

Profit – the income remaining after all cost

BREAK-EVEN ANALYSIS

$$D_{BEP} = \frac{FC}{p - VC}$$

Where:

R=revenue

VC_T = total variable cost

D_{BEP} = (Break-even point), quantity sold

BREAK-EVEN ANALYSIS

A farmer produces bottled tomato sauce. The fixed costs (equipment, rent, etc.) are ₱50,000 per season. Each bottle sells for ₱100, and the variable cost (ingredients, packaging) is ₱40 per bottle. How many bottles must be sold to break even?

BREAK-EVEN ANALYSIS

A small shop in Bulacan fabricates threshers for palay producers in the locality. The shop can produce each thresher at a labor cost of P1,800. The cost of materials for each unit is P2,500. The variable costs amount to P650 per unit, while fixed charges incurred per annum total P69,000. If the portable threshers are sold for P7,800 per unit, how many units must be produced and sold per annum to break-even?

BREAK-EVEN ANALYSIS

- **Break-even Point (of two alternatives)** - Value of the common variable of two alternatives that will result in equal cost for the alternatives considered

PROFIT MAXIMIZATION

$$P = a - bD$$

Demand that will maximize profit: $D^* = \frac{a - VC}{2b}$

Demand that will maximize revenue: $D' = \frac{a}{2b}$

BREAK-EVEN ANALYSIS

In producing farm tillers, Kasaka Ltd. Has determined that its selling price (p) and the quantity it sell (D) per month are related by the equation $p = 1350 - 0.15D$. If the fixed costs per month amount to Php200,000 and the variable cost per unit is Php300, determine the number of units that will maximize net profit and the maximum net profit.

BREAK-EVEN ANALYSIS

Magma Tierra is engaged in the research, design, and manufacturing of farm implements. The company estimates that the relationship between unit price and demand per month for a potential product is approximated by $p = 112 - 0.01D$. Fixed costs are estimated to be Php72,000 per month, and the estimated variable costs are Php26 per unit. How much is the maximum profit that the company can realize per month?

DEPRECIATION

- is the decrease in the value of physical property with the passage of time.

Purpose of Determining Depreciation:

- To provide for the recovery of capital which has been invested in physical property.
- To enable the cost of depreciation to be charged to the cost of producing products or services that results from the use of the property.

DEPRECIATION

A property is depreciable if it meets all three requirements as follows:

1. It must be used in business or held for the production of income.
2. It must have determinable life, and the life must be longer than one year.
3. It must be something that wears out, decays, gets used up, become obsolete, or losses value from natural cause.

DEPRECIATION

TERMS

- Depreciation Cost – depends upon the physical or economic life of the equipment and its first cost.
- Physical Life – the length of time during which it is capable of performing the function for which it was designed and manufactured.
- Economic Life – the length of time during which it will operate at a satisfactory profit.
- Useful Life – the period of time as asset is kept in productive use in a trade or business.

DEPRECIATION

TERMS

- Value – present worth of all future profits that are to be received through the ownership of a property.
- First Cost – includes the original purchase price, freight and transportation charges to the site, installation expenses, initial taxes and permits to operate, and all other expenses needed to put the equipment into operation.
- Salvage Value – second-hand value or resale value; the amount for which the equipment or machine can be sold as second-hand.
- Scrap or Junk Value – amount the equipment can be sold for, when disposed of as a junk.

DEPRECIATION

TERMS

- Market Value – the amount which a willing buyer will pay to a willing seller for a property, where each has equal advantage and is under no compulsion to buy or sell.
- Utility or Use Value – is what the property is worth to the owner as an operating unit.
- Fair Value – is usually determined by a disinterested third party in order to establish a price that is fair to both seller and buyer.
- Book Value/ Depreciated Book Value – is the worth of a property as shown on the accounting records of an enterprise.

DEPRECIATION

TERMS

- Going Value or Going-Concern Value – considered to be an intangible value, which an actually operating concern has due to its operation.
- Goodwill Value – that element of value, which a business has earned through the favorable consideration and patronage of its customers arising from its well-known and well-conducted policies and operations.
- Rate-Base Value – the value assigned to the property for the purpose of establishing rates.
- Franchise Value – is an intangible item of value arising from the exclusive right of the company to provide a specific product or service in a stated region of the country.

Straight-line Method

The loss on value is directly proportional to the age of the equipment or asset; simplest method; gives uniform annual charge.

$$d = \frac{C_o - C_n}{n} \quad D_m = md = (C_o - C_n) \frac{m}{n} \quad C_m = C_o - D_m$$

n = useful life of the property in years

m = age of the property at any time less than or equal to n

d = annual cost of depreciation

D_m = accrued or total depreciation up to m years

C_o = original or first cost of the property

C_m = book value of the property at the end of m years

C_n = book value at the end of life, n , years [salvage value or scrap value]

C_n = book value at the end of n years

Sample Problem

If an asset has first cost of Php50 000 with a Php1 ,000 estimated salvage value after 5 years, (a) calculate the annual depreciation and (b) calculate the book value of the asset after each year, using straight line depreciation.

Sample Problem

An asset that is book-depreciated over a 5-year period by the straight-line method has $C_3 = \text{Php}62,000$ with a depreciation charge of $\text{Php}26,000$ per year. Determine (a) the first cost of the asset and (b) the assumed salvage value.

Sample Problem

A tax and duty-free importation of a 30-Hp sand mill (for paint manufacturing) cost Php360 000. Bank charges and brokerage cost P5,000. Foundation and installation costs were Php25 000. Other incidental expenses amounted to Php20 000. Salvage value of the mill is estimated to be Php60 000 after 20 years. Find the appraisal value of the mill at the end of (a) 10 years and (b) 15 years.



Sinking Fund Method

The sinking fund is established in which funds will accumulate for replacement purposes.

The sinking fund method of depreciation is a method where funds are set aside annually to accumulate and cover the cost of replacing an asset at the end of its useful life, with the accumulated funds earning interest.



Sinking Fund Method

$$d = (C_o - C_n) \frac{1}{\left(\frac{F}{A}, i\%, n\right)} \quad D_m = (C_o - C_n) \frac{\left(\frac{F}{A}, i\%, m\right)}{\left(\frac{F}{A}, i\%, n\right)}$$

$$C_m = C_o - D_m$$

n = useful life of the property in years

m = age of the property at any time less than or equal to n

d = annual cost of depreciation

D_m = accrued or total depreciation up to m years

C_o = original or first cost of the property

C_m = book value of the property at the end of m years

C_n = book value at the end of life, n, years [salvage value or scrap value]

C_n = book value at the end of n years



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Sample Problem

An asset has a first cost of Php5,000 and an estimated salvage value of Php1,000 at the end of four years. Determine the annual depreciation using (a) straight-line method and, (b) Sinking-fund Method at 10%.



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Sample Problem

A drill press is purchased for Php1,100 and has an expected life of 12 year, at which the salvage value is estimated to be Php130. What will be the book value at the end of 8 years by (a) Straight-line method and (b) 4% sinking fund depreciation?



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Sample Problem

The original cost of a certain piece of equipment is Php150 000 and it is depreciated by a 10% sinking fund method. Determine the annual depreciation charge if the book value of the equipment after 10 years is the same as if it had been depreciated at Php 14 000 each year by straight-line formula.



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Sample Problem

An asset was purchased 10 years ago for Php2,400. It is being depreciated in accordance with straight-line method for an estimated life of 20 years and salvage value of Php400. What is the difference in its book value and the book value that would have resulted if 6% sinking-fund depreciation has been used?

Declining Balance Method

The annual cost of depreciation is a fixed percentage of the book value at the beginning of the year; also called the Matheson Formula and Constant Percentage Method.

$$k = 1 - \sqrt[m]{\frac{C_m}{C_o}} = 1 - \sqrt[n]{\frac{C_n}{C_o}} \qquad C_m = C_o(1 - k)^m$$
$$d_m = kC_{m-1} \qquad C_n = C_o(1 - k)^n$$

DEPRECIATION

The cost of a certain asset is P3,000; its life is 6 years and scrap value is P500. Using Declining Balance Method, find the annual rate of percentage.

Sum of Years' Digits (SYD) Method

The Sum of Years' Digits (SYD) method is an accelerated depreciation method used in accounting. It results in higher depreciation expenses in the early years of an asset's life and lower expenses in later years, reflecting the idea that assets are typically more productive when they are newer.

$n * (n + 1) / 2$, where 'n' is the useful life

Annual Depreciation Expense: (Remaining useful life / Sum of the years' digits) * Depreciable Basis.

DEPRECIATION

A company purchases an asset for P10,000.00 and plans to keep it for 20 years. If the salvage value is zero at the end of the 20th year, what is the depreciation in the 3rd year by Sum-of-the-Years Digit method?

DEPRECIATION

If an asset costs Php10,000, has a salvage value of Php1,000, and a useful life of 5 years: what is the depreciation each year by Sum-of-the-Years Digit method?

DEPRECIATION

A civil engineer bought a gantry crane for erecting tall buildings. It was invoiced from Japan CIF Manila at Php250, 000.00. Brokerage, bank, arrastre fees, customs' duties, permits, etc. total Php 120,000.00. At the end 10 years, he expects to sell it for Php 50, 000.00. Calculate the depreciation using SYD.

The benefits to society regardless of who in the society receives the benefits.

- A. Financial benefit
- B. Economic benefit
- C. Revenue
- D. Sales

The benefits to society regardless of who in the society receives the benefits.

- A. Financial benefit
- B. Economic benefit**
- C. Revenue
- D. Sales

An **economic benefit** refers to the overall advantage or gain to society from an activity, investment, or policy, regardless of who specifically receives the benefit. It includes improvements in well-being, increased efficiency, job creation, and other positive outcomes that contribute to the overall economy and public welfare.

Financial benefit usually refers to gains in monetary terms to specific individuals or organizations.

Revenue is the income generated by a business from its normal activities.

Sales refer to the exchange of goods or services for money.

Benefits less costs excluding depreciation.

- A. Profit
- B. Net benefit
- C. Cash flow
- D. Economic benefits

Benefits less costs excluding depreciation.

A. Profit

B. Net benefit

C. Cash flow

D. Economic benefits

Which one is a one-time expense?

- A. Investment
- B. Single cost
- C. One-time cost
- D. Operating cost

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A. Investment

B. Single cost

C. One-time cost

D. Operating cost

It refers to an annually recurring cost.

- A. Investment
- B. Operating cost
- C. Contingency
- D. Operating capital

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A. Investment

B. Operating cost

C. Contingency

D. Operating capital

Land is an example of

- A. Investment
- B. Operating cost
- C. Contingency
- D. Operating capital

Land is an example of

A. Investment

B. Operating cost

C. Contingency

D. Operating capital

Manager's salary is an example of

- A. Investment
- B. Operating cost
- C. Contingency
- D. Operating capital

Manager's salary is an example of

A. Investment

B. Operating cost

C. Contingency

D. Operating capital

Contingency is normally what percentage of investment cost?

- A. 5%
- B. 10%
- C. 15%
- D. 20%

Contingency is normally what percentage of investment cost?

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B. 10%

C. 15%

D. 20%

What refers to the scrap value of a property?

- A. Salvage value
- B. Economic value
- C. Book value
- D. End value

What refers to the scrap value of a property?

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- B. Economic value
- C. Book value
- D. End value

Using Straight Line Method, what is the annual depreciation of a tractor purchased at Php 2,500,000 if its life span is assumed at 8 years? Salvage value is 10%.

- | | |
|----------------|----------------|
| A. Php 218,250 | B. Php 281,250 |
| C. Php 821,250 | D. Php 250,281 |

Using Straight Line Method, what is the annual depreciation of a tractor purchased at Php 2,500,000 if its life span is assumed at 8 years? Salvage value is 10%.

- A. Php 218,250 B. Php 281,250
C. Php 821,250 D. Php 250,281

What feasibility indicator is best for determining how fast the project recovers the investment?

A. NPV

B. IRR

C. BCR

D. Payback period

What feasibility indicator is best for determining how fast the project recovers the investment?

A. NPV

B. IRR

C. BCR

D. Payback period

What feasibility indicator is best for determining how much can you gain from a certain project?

A. NPV

B. IRR

C. BCR

D. Payback period

What feasibility indicator is best for determining how much can you gain from a certain project?

A. NPV

B. IRR

C. BCR

D. Payback period

What feasibility indicator is normally compared to bank's interest rate?

A. NPV

B. IRR

C. BCR

D. Payback period

What feasibility indicator is normally compared to bank's interest rate?

A. NPV

B. IRR

C. BCR

D. Payback period

Which among the feasibility indicators is best for determining project performance?

A. NPV

B. IRR

C. BCR

D. Payback period

Which among the feasibility indicators is best for determining project performance?

A. NPV

B. IRR

C. BCR

D. Payback period

The discount rate when the Net Present Value is zero.

- A. Break-even rate
- B. IRR
- C. BCR
- D. ROI

The discount rate when the Net Present Value is zero.

A. Break-even rate

B. IRR

C. BCR

D. ROI

At project start, the net cash flow of the cassava processing venture is -7 million pesos due to loan and revenue is not materialized yet. Discount rate is 10%. What is the value of the discounted cash flow in million pesos at project start?

- A. 7
- B. -7
- C. -7.07
- D. -7.7

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B. -7

C. -7.07

D. -7.7

Using financial feasibility analysis, the Net Present Values (NPVs) of farm-to-market road, irrigation and flood control projects are 22.5, 7.5 and -3.1 million pesos, while the Internal Rate of Returns (IRRs) are 15%, 7% and 2.1%, respectively. Annual discount plus inflation rate is 10%. Which project shall be implemented?

- A. Farm-to-market road
- B. Irrigation
- C. Flood control
- D. None

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A. Farm-to-market road

B. Irrigation

C. Flood control

D. None

Using straight line depreciation method, what is the book value at the end of the third year of a tractor purchased at Php 2,500,000 if its life span is assumed at 10 years? Salvage value is 10%.

- A. Php1,285,000
- B. Php1,582,000
- C. Php1,825,000
- D. Php1,852,000

Using straight line depreciation method, what is the book value at the end of the third year of a tractor purchased at Php 2,500,000 if its life span is assumed at 10 years? Salvage value is 10%.

A. Php1,285,000

B. Php1,582,000

C. Php1,825,000

D. Php1,852,000

Using economic feasibility analysis, the Net Present Values (NPVs) of small water impounding project (SWIP), shallow tubewell, small scale irrigation project (SSIP) are -0.5, -0.9 and 0 million pesos, respectively. The proposed projects are owned by the government. Which project is economically feasible?

- A. None
- B. SWIP
- C. Shallow tube well
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Microhydro, solar electrification and covered lagoon biodigester projects are evaluated for financial feasibility. Each requires PHP 20,000,000 investment. Only one project can be funded. The financial internal rates of return are 15%, 9% and 8%, respectively. Annual discount plus inflation rate is 10%. Which project shall be implemented?

- A. Covered lagoon biodigester
- B. Microhydro
- C. Solar electrification
- D. None

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A banana production and export project is being evaluated for financial feasibility. At 12% annual discount rate, it has the following feasibility indicators: net present value of 1.8 billion pesos and internal rate of return of 58%. If annual discount rate drops to 9%, will it be feasible?

- A. No
- B. Yes
- C. Maybe
- D. Insufficient data to determine

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B. Yes

C. Maybe

D. Insufficient data to determine

 *Key principle:*

If a project is feasible at a certain discount rate, it will be even more feasible at a lower discount rate, assuming all other factors remain the same.